

SQL Basics Assignment

Q1. Create a New Database and Table for Employees

Task: Create a new database named company_db and create a table named employees with the following columns:

Column Name	Data Type	Constraint
employee_id	INT	PRIMARY KEY
first_name	VARCHAR(50)	
last_name	VARCHAR(50)	
department	VARCHAR(50)	
salary	INT	
hire_date	DATE	

Answer- The SQL Commands/Queries we use to Complete this task are

```
CREATE DATABASE company_db;          (To Create Database)
```

```
USE company_db;    (To use the created Database)
```

```
CREATE TABLE employees (  
    employee_id INT      PRIMARY KEY,  
    first_name  VARCHAR(50),  
    last_name   VARCHAR(50),  
    department  VARCHAR(50),  
    salary      INT,  
    hire_date   DATE  
);
```

```
-- Check the table was created
```

```
SHOW TABLES;
```

```
-- Check the column structure
```

```
DESCRIBE employees;
```

The Result Grid

Field	Type	Null	Key	Default	Extra
Employee_ID	int	NO	PRI	NULL	
First_name	varchar(50)	YES		NULL	
Last_name	varchar(50)	YES		NULL	
Department	varchar(50)	YES		NULL	
Salary	int	YES		NULL	
Hire_date	date	YES		NULL	

Q2. Insert Data into Employees Table Task: Insert the following sample records into the employees' table

employee_id	first_name	last_name	department	salary	hire_date
101	Amit	Sharma	HR	50000	2020-01-15
102	Riya	Kapoor	Sales	75000	2019-03-22
103	Raj	Mehta	IT	90000	2018-07-11
104	Neha	Verma	IT	85000	2021-09-01
105	Arjun	Singh	Finance	60000	2022-02-10

Answer-

The SQL Commands/Queries we Used to complete this given Task are:

INSERT INTO employees (employee_id, first_name, last_name, department, salary, hire_date)

VALUES

(101, 'Amit', 'Sharma', 'HR', 50000, '2020-01-15'),

(102, 'Riya', 'Kapoor', 'Sales', 75000, '2019-03-22'),

(103, 'Raj', 'Mehta', 'IT', 90000, '2018-07-11'),

(104, 'Neha', 'Verma', 'IT', 85000, '2021-09-01'),

(105, 'Arjun', 'Singh', 'Finance', 60000, '2022-02-10');

And then verified by

SELECT * FROM employees;

Employee_ID	First_name	Last_name	Department	Salary	Hire_date
101	Amit	Sharma	HR	50000	2020-01-15
102	Riya	Kapoor	Sales	75000	2019-03-22
103	Raj	Mehta	IT	90000	2018-07-11
104	Neha	Verma	IT	85000	2021-09-01
105	Arjun	Singh	Finance	60000	2022-02-10
NULL	NULL	NULL	NULL	NULL	NULL

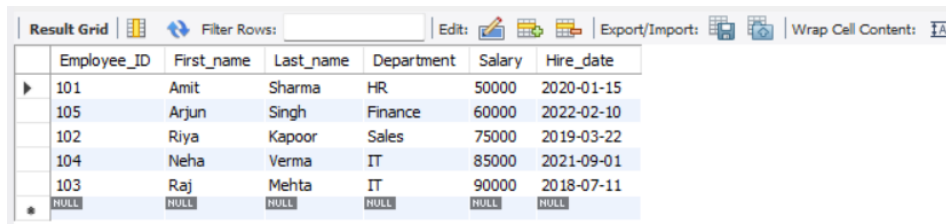
Q3. Display All Employee Records Sorted by Salary (Lowest to Highest) Hint: Use the ORDER BY clause on the salary column.

Answer- The SQL Commands/Queries we Used to complete this given Task are:

```
SELECT * FROM employees
```

```
ORDER BY salary ;
```

Result Grid

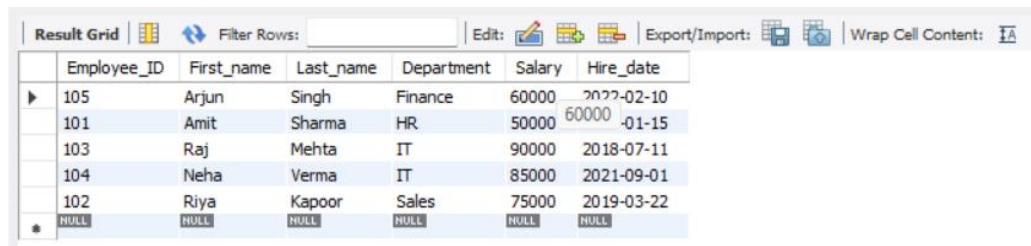


Employee_ID	First_name	Last_name	Department	Salary	Hire_date
101	Amit	Sharma	HR	50000	2020-01-15
105	Arjun	Singh	Finance	60000	2022-02-10
102	Riya	Kapoor	Sales	75000	2019-03-22
104	Neha	Verma	IT	85000	2021-09-01
103	Raj	Mehta	IT	90000	2018-07-11
NULL	NULL	NULL	NULL	NULL	NULL

Q4. Show Employees Sorted by Department (A–Z) and Salary (High → Low)

Answer- The SQL Commands/Queries we Used to complete this given Task are:

```
select * From employees Order BY Department asc, Salary desc;
```

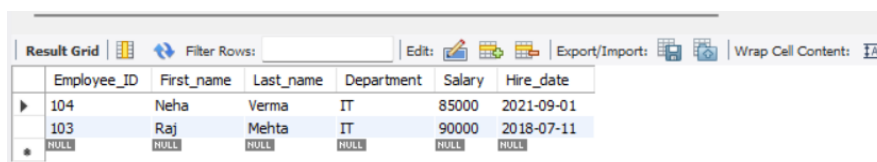


Employee_ID	First_name	Last_name	Department	Salary	Hire_date
105	Arjun	Singh	Finance	60000	2022-02-10
101	Amit	Sharma	HR	50000	2020-01-15
103	Raj	Mehta	IT	90000	2018-07-11
104	Neha	Verma	IT	85000	2021-09-01
102	Riya	Kapoor	Sales	75000	2019-03-22
NULL	NULL	NULL	NULL	NULL	NULL

Q5. List All Employees in the IT Department, Ordered by Hire Date (Newest First)

Answer- The SQL Commands/Queries we Used to complete this given Task are:

```
Select * from employees where Department = 'IT' Order BY Hire_date desc;
```



Employee_ID	First_name	Last_name	Department	Salary	Hire_date
104	Neha	Verma	IT	85000	2021-09-01
103	Raj	Mehta	IT	90000	2018-07-11
NULL	NULL	NULL	NULL	NULL	NULL

Q6. Create and Populate a Sales Table Task: Create a table sale to track sales data:

sale_id	customer_name	amount	sale_date
1	Aditi	1500	2024-08-01
2	Rohan	2200	2024-08-03
3	Aditi	3500	2024-09-05
4	Meena	2700	2024-09-15
5	Rohan	4500	2024-09-25

Answer- The SQL Commands/Queries we Used to complete this given Task are:

```
USE company_db;
```

```
CREATE TABLE sales (
```

```
    sale_id    INT,
```

```
    customer_name VARCHAR(100),
```

```
    amount     INT,
```

```
    sale_date  DATE
```

```
);
```

```
INSERT INTO sales (sale_id, customer_name, amount, sale_date) VALUES
```

```
(1, 'Aditi', 1500, '2024-08-01'),
```

```
(2, 'Rohan', 2200, '2024-08-03'),
```

```
(3, 'Aditi', 3500, '2024-09-05'),
```

```
(4, 'Meena', 2700, '2024-09-15'),
```

```
(5, 'Rohan', 4500, '2024-09-25');
```

```
SELECT * FROM sales;
```

Result Grid

Filter Rows:

Export:

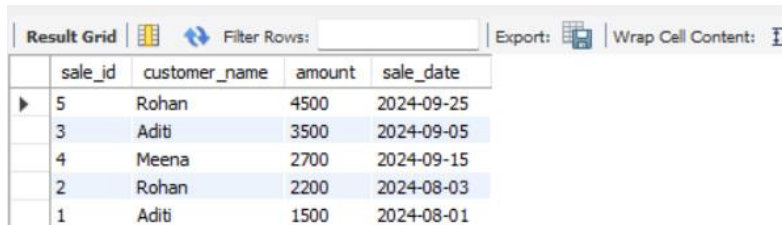
Wrap Cell Content:

	sale_id	customer_name	amount	sale_date
▶	1	Aditi	1500	2024-08-01
	2	Rohan	2200	2024-08-03
	3	Aditi	3500	2024-09-05
	4	Meena	2700	2024-09-15
	5	Rohan	4500	2024-09-25

Q7. Display All Sales Records Sorted by Amount (Highest → Lowest) Hint: Use ORDER BY amount DESC.

Answer- The SQL Commands/Queries we Used to complete this given Task are:

Select * from sales order by amount desc;



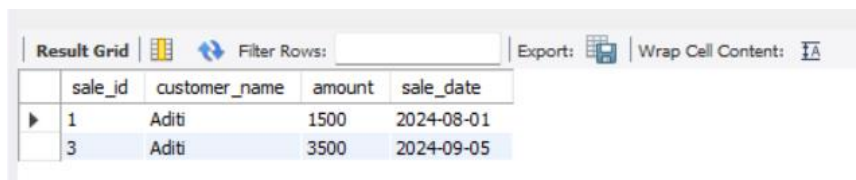
The screenshot shows a database interface with a 'Result Grid' tab. The grid displays five rows of sales data, sorted by 'amount' in descending order. The columns are 'sale_id', 'customer_name', 'amount', and 'sale_date'. The rows are: (5, Rohan, 4500, 2024-09-25), (3, Aditi, 3500, 2024-09-05), (4, Meena, 2700, 2024-09-15), (2, Rohan, 2200, 2024-08-03), and (1, Aditi, 1500, 2024-08-01).

	sale_id	customer_name	amount	sale_date
▶	5	Rohan	4500	2024-09-25
	3	Aditi	3500	2024-09-05
	4	Meena	2700	2024-09-15
	2	Rohan	2200	2024-08-03
	1	Aditi	1500	2024-08-01

Q8. Show All Sales Made by Customer “Aditi” Hint: Use WHERE customer_name = 'Aditi'.

Answer- The SQL Commands/Queries we Used to complete this given Task are:

Select * from sales where customer_name='Aditi';



The screenshot shows a database interface with a 'Result Grid' tab. The grid displays two rows of sales data for the customer 'Aditi'. The columns are 'sale_id', 'customer_name', 'amount', and 'sale_date'. The rows are: (1, Aditi, 1500, 2024-08-01) and (3, Aditi, 3500, 2024-09-05).

	sale_id	customer_name	amount	sale_date
▶	1	Aditi	1500	2024-08-01
	3	Aditi	3500	2024-09-05

Q9. What is the Difference Between a Primary Key and a Foreign Key?

Answer-

Feature	Primary Key	Foreign Key
Purpose	Uniquely identify a row	Link two tables
Null Allowed	No	Yes
Duplicates	No	Yes
Per Table	Only One	Can have many
Define in	Same Table	Reference Another Table

Q10. What Are Constraints in SQL and Why Are They Used?

Answer-

Definition: Constraints are rules applied to columns in a table to ensure that the data stored is accurate, valid, and reliable. They are defined at the time of table creation or can be added later using the ALTER TABLE command.

Types of Constraints:

1. PRIMARY KEY

Uniquely identifies each record in a table. It does not allow NULL or duplicate values.

SQL Query : sale_id INT PRIMARY KEY

2. NOT NULL

Ensures that a column cannot have an empty or NULL value.

SQL Query : `customer_name VARCHAR(100) NOT NULL`

3. UNIQUE

Ensures all values in a column are different from each other.

SQL Query : `email VARCHAR(100) UNIQUE`

4. DEFAULT

Automatically assigns a default value to a column if no value is provided.

SQL Query : `sale_date DATE DEFAULT '2024-01-01'`

5. CHECK

Ensures that values in a column satisfy a specific condition.

SQL Query : `amount INT CHECK (amount > 0)`

6. FOREIGN KEY Links a column in one table to the primary key of another table, establishing a relationship between them.

SQL Query : `FOREIGN KEY (customer_id) REFERENCES customers(customer_id)`

Why Are Constraints Used?

- To maintain data integrity and accuracy
- To prevent invalid or incomplete data from being entered
- To enforce relationships between tables
- To make the database consistent and reliable